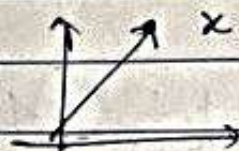


24th Feb 2026 / DA6360

⇒ Unconstrained Optimizaⁿ in $\mathbb{R}^d$

$$\min_{x \in \mathbb{R}^d} f(x)$$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

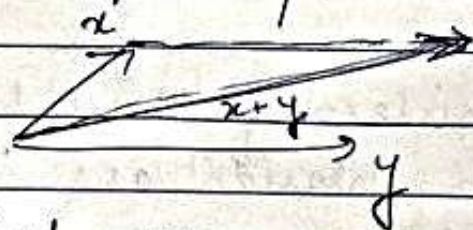


(pt.)
- Norm of a vector $\rightarrow$
 $(\|x\|) \Rightarrow$ (distance from the origin)

- Positivity : $\|x\| = 0 \Rightarrow x = 0$

- scaling : $\|tx\| = |t| \|x\|$, t is a scalar no. i.e. a scalar

- Triangle Inequality : $\|x+y\| \leq \|x\| + \|y\|$



- some useful norms

1. l_2 $\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_d^2}$
(Euclidean norm) $= \sqrt{x^T x}$

2. l_1 $\|x\|_1 = \sum_{i=1}^d |x_i|$
(Manhattan dist.)

↪ kind of total steps taken.

$$3. \quad l_p \quad \|x\|_p = \left(|x_1|^p + |x_2|^p + \dots + |x_d|^p \right)^{1/p}$$

as p increases, it tends to focus on the maximum value.

$$4. \quad l_\infty \quad \|x\|_\infty = \max_{i=1 \dots d} |x_i|$$

→ norm induces distance

$$d(x, y) = \|x - y\|$$

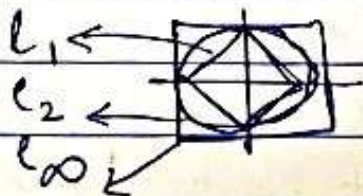
$$x, y \in \mathbb{R}^d$$

- Norm ball

$$B_p(\hat{x}, r) = \left\{ x : \|x - \hat{x}\|_p \leq r \right\}$$

ball around pt. of rad.

- Unit ball



balls are contained within one another

$$B_1(0, 1) \subseteq B_2(0, 1) \subseteq \dots \subseteq B_\infty(0, 1)$$

- Equivalence of Norms
pick any $p \neq q$

$$0 < c_{\text{low}} < c_{\text{high}}$$

$$c_{\text{low}} \|x\|_q \leq \|x\|_p \leq c_{\text{high}} \|x\|_q$$

one is perhaps straightforward always

- Weighted L_2 Norm -

$\|x\|_M$, M is a diagonal matrix with +ve entries.

$$\|x\|_M = \sqrt{x^T M x}$$
$$= \sqrt{x_1^2 M_1 + x_2^2 M_2 + x_d^2 M_d}$$

⇒ Representaⁿ of pts./vectors in R^d

- we want to tell where each pt. resides in the space

$V = \{v_1, \dots, v_d\}$ is a basis

- pick $x \in R^d$, starting from origin how to reach x using $\underbrace{v_1, \dots, v_d}_{\text{direc^{ns}}}$

$$x = \gamma_1 v_1 + \gamma_2 v_2 + \gamma_3 v_3 + \dots + \gamma_d v_d$$

- spanning property: For every $x \in R^d$, we should be able to express as

$$x = \gamma_1 v_1 + \gamma_2 v_2 + \dots + \gamma_d v_d, \gamma_i \in R$$

$i = 1, \dots, d$

- Redundancy - any direcⁿ v_i expressible in terms of other direc^{ns} is said to be redundant.

$$v_i = \gamma_1 v_1 + \dots + \gamma_d v_d$$

↳ then this is redundant,

& v_i is linearly dependent on other direc^{ns}.

- No redundancy: $\sum_{i=1}^d \gamma_i x_i = 0$.

$\Rightarrow \gamma_i = 0, \forall i = 1, \dots, d$

case I: $\gamma_1 > 0, \gamma_2, \dots, \gamma_d = 0, \gamma_1 v_1 = 0$

$\Rightarrow v_1 = 0$ (vacuous)

case II: $> 1 \gamma_i > 0$

$\gamma_{i_1} v_{i_1} + \gamma_{i_2} v_{i_2} = 0$

$\Rightarrow \cancel{\gamma_{i_1} v_{i_1}} + \cancel{\gamma_{i_2} v_{i_2}} = 0$

$\Rightarrow \gamma_{i_1} v_{i_1} = -\gamma_{i_2} v_{i_2}$

one is dependent on the other 2.

- Std. Basis

$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \rightarrow i^{\text{th}} \text{ pt.}$

$x = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix} = x_1 e_1 + \dots + x_d e_d$

Given a basis $V = \{v_1, \dots, v_d\}$,

$x \in \mathbb{R}^d$. Find out how to represent x in basis V .

$x_V = \begin{bmatrix} x_{v_1} \\ \vdots \\ x_{v_d} \end{bmatrix}$

$$x_{u_1} u_1 + \dots + x_{u_d} u_d = x$$

$$\begin{bmatrix} | & & | \\ u_1 & \dots & u_d \\ | & & | \end{bmatrix} \begin{bmatrix} x_{u_1} \\ \vdots \\ x_{u_d} \end{bmatrix} = x$$

$$\underbrace{\begin{bmatrix} | & & | \\ u_1 & \dots & u_d \\ | & & | \end{bmatrix}}_V \underbrace{\begin{bmatrix} x_{u_1} \\ \vdots \\ x_{u_d} \end{bmatrix}}_{x_u = V^T x} = x$$

- Dot Product

$$\langle u_1, u_2 \rangle = u_1^T u_2 = \sum_{i=1}^d u_{1i} u_{2i}$$

$$= u_{11} u_{21} + \dots + u_{1d} u_{2d}$$

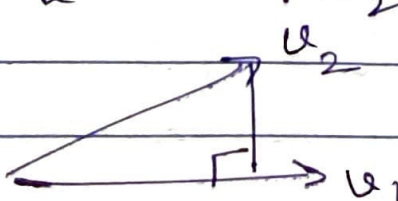
two vectors are orthogonal if $\langle u_1, u_2 \rangle = 0$

scalar

- Best possible approximation of u_2 in terms of u_1 is given by

$$\left[\langle \frac{u_1}{\|u_1\|}, u_2 \rangle \frac{u_1}{\|u_1\|} \right]$$

by $\min_t \|u_2 - t u_1\|_2^2$ (for proof)



- Orthogonal Basis

$$V = \{v_1, \dots, v_d\} \text{ is a basis}$$

$$\langle v_i, v_j \rangle = 0, \quad \forall i \neq j$$

- Orthonormal Basis

orthogonal + $\|v_i\| = 1$

eg. in 2d

$$V = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$$V = \left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}, \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \right\}$$

$\Rightarrow$ Linear Transformⁿ

- Transformⁿ is linear

$$(1) \quad T(\gamma x) = \gamma T(x)$$

$$(2) \quad T(x_1 + x_2) = T(x_1) + T(x_2)$$

$$\begin{aligned} \Rightarrow T(\gamma_1 x_1 + \gamma_2 x_2) &= T(\gamma_1 x_1) + T(\gamma_2 x_2) \\ &= \gamma_1 T(x_1) + \gamma_2 T(x_2) \end{aligned}$$

- enough to understand how the Transformⁿ acts on the basis

$$e_1 \longrightarrow T e_1$$

$$\vdots$$

$$e_d \longrightarrow T e_d$$

$$x = x_1 e_1 + \dots + x_d e_d$$

$$\downarrow T$$

$$x_1 T(e_1) + \dots + x_d T(e_d)$$

The matrix that represents transformation T is

$$M_T = \begin{bmatrix} | & & | \\ T(e_1) & \dots & T(e_d) \\ | & & | \end{bmatrix}$$

eg.

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & -1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \xrightarrow{T} \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$